

Week 4 Reflection
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Motion Planning has given me a deeper understanding of the application of inverse kinematics, specifically the Jacobian approach, which I had previously assumed was not something applicable to this project. As I started building the code for servo communication, I saw in the registers that a given position was available for speed (reading current and writing desired). I didn't think much of this at first. I figured that giving an arm a position to move to was enough for teleoperation, which it is. However, for smooth operation and to minimize position hunting during quicker movements, translating the speed between servos would be essential. This is the heart of the Jacobian approach.

For simplicity, basic geometric and algebraic approaches work well with what I have set up, even if it is more cumbersome computationally. With future approaches, however, including those that require maneuvering around complex and changing environments, and decision-making with AI models paired with vision, it will be necessary to apply the Jacobian methods of inverse kinematics.

Everything up to this point has been about systems of control or the 'how' of motion. Planning is the next layer above control. Two major planning models are common in robotics: Rapidly-Exploring Random Trees (RRT) and Probabilistic Roadmap (PRM). The RRT has several variants, but overall, the process uses a dimensional space for Collision Free (C-free), where the arm is 'able' to move and samples points in space randomly; each sample could lead to a collision or not. This sampling of the C-free occurs first at the base, then works towards the end effector. If collisions occur repeatedly at any joint, then the previous joint has its position resampled, and the process repeats.

An optimization for this is to sample locations near previously sampled ones and direct smaller changes in positions during planning. One of the most widely used implementations of RRT includes Bi-directional RRT, where instead of the base being joint 0, there are two bases, the previously mentioned joint 0 and the end effector. The idea is that sampling from each end randomly in the C-free means that they should find a common path building in the same direction toward the midpoint between them.

Probabilistic Roadmap (PRM) uses a different approach; instead of taking a solution and sequentially sampling probabilities to find appropriate paths forward, PRM uses pre-arranged roadmaps through the C-free dimension and verified C-free edges that dictate movement forward. This requires an upfront cost of developing the roadmap of available paths of motion. The PRM is ideal for static and planned environments and suffers in dynamic settings. The plus side to using PRM approaches is that it requires less computational overhead when compared to even optimized RRT approaches.

Given how these approaches vary, I can see how PRM has its benefits, but I think for future implementations, especially for pairing with AI vision, I will have to utilize RRT. Specifically using the Bidirectional approach. Unfortunately, these implementations of motion planning will be saved for another directed reading.